3. Operating Modes

3.1 Operating Mode Types and Selection

There are two types of operating-mode selection: one is be selected by the level on pins at the time of release from the reset state, and the other is selected by software after release from the reset state.

Table 3.1 lists the relationship between levels on the mode-setting pins (MD and UB) on release from the reset state and the operating mode selected at that time. For details on each of the operating modes, refer to section 3.3, Details of Operating Modes. Operation starts with the on-chip ROM (code flash memory and data flash memory) enabled and the external bus disabled, regardless of the mode in which operation started. Set the SYSCR0.EXBE bit to 1 (external bus enabled) to enable the external bus.

Table 3.1 Selection of Operating Modes by the Mode-Setting Pins on Release from the Reset State

Mode-Setting Pin		Operating Mode	SYSCR0 Initial State	
MD*1	UB*2		ROME	EXBE
High	—	Single-chip mode	1 (On-chip ROM enabled)	0 (External bus disabled)
Low	Low	Boot mode (SCI interface)		
	High	Boot mode (USB interface)		
Low → High*3	Low	Boot mode (FINE interface)		

Note 1. Do not change the level on the MD pin while the MCU is operating.
Note 2. The PC7 pin, which is multiplexed with the UB pin function, may also be used as a general port pin or as an input or output pin for peripheral functions.
Note 3. After release the reset state while the MD pin is at the low level, switch it to the high level within 20 to 100 msec.

Table 3.2 gives a list of the operating mode settings that can be made with system control register 0 (SYSCR0). For details on each of the operating modes, refer to section 3.3, Details of Operating Modes.

Table 3.2 Selection of Operating Modes by Register Setting

SYSCR0		Operating Mode
ROME	EXBE	
0 (On-chip ROM disabled)*1	0 (external bus disabled)	Single-chip mode
1 (On-chip ROM enabled)	0 (external bus disabled)	
0 (On-chip ROM disabled)*1	1 (external bus enabled)	On-chip ROM disabled extended mode
1 (On-chip ROM enabled)	1 (external bus enabled)	On-chip ROM enabled extended mode

Note 1. Once the ROME bit is set to 0, it cannot be reverted to 1.

The endian is selectable in single-chip mode. Endian is selected by the endian selection bits (MDE[2:0]) in the endian select register (MDE). Table 3.3 lists the correspondence between the setting and endian. For details on selection of endian, refer to section 7.2.5, Endian Select Register (MDE).

Table 3.3 Selection of Endian

Setting of the MDE[2:0] Bits	Selected Endian
000b	Big endian
111b	Little endian

3.2 Register Descriptions

3.2.1 Mode Monitor Register (MDMONR)

Address(es): 0008 0000h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	MD
Value after reset:	0	0	0	0	0	0	0	x	0	0	0	0	0	0	0	0/1*1

Bit	Symbol	Bit Name	Description	R/W
b0	MD	MD Pin Status Flag	0: The MD pin is low. 1: The MD pin is high.	R
b7 to b1	—	Reserved	These bits are read as 0.	R
b8	—	Reserved	The read value is undefined.	R
b15 to b9	—	Reserved	These bits are read as 0.	R

Note 1. This affects the level on the MD pin at the time of release from the reset state.

3.2.2 System Control Register 0 (SYSCR0)

Address(es): 0008 0006h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	KEY[7:0]								—	—	—	—	—	—	EXBE	ROME
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit	Symbol	Bit Name	Description	R/W
b0	ROME	On-Chip ROM Enable	0: The on-chip ROM is disabled. 1: The on-chip ROM is enabled.	R/W
b1	EXBE	External Bus Enable	0: The external bus is disabled. 1: The external bus is enabled.	R/W
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b15 to b8	KEY[7:0]	SYSCR0 Key Code	These bits control permission and prohibition of writing to the SYSCR0 register. To modify the SYSCR0 register, write 5Ah to the eight higher-order bits and the desired value to the eight lower-order bits as a 16-bit unit.	R/(W)*1

Note 1. Write data is not retained.

ROME Bit (On-Chip ROM Enable)

The ROME bit enables or disables the on-chip ROM (code flash memory and data flash memory).

Once set to 0, it cannot be reverted to 1.

A 0 should not be written to this bit while a program is being executed from the on-chip ROM (code flash memory and data flash memory). After writing a 0 to this bit, be sure to disable the on-chip ROM by changing the ROME bit to 0 before proceeding with further processing.

EXBE Bit (External Bus Enable)

The EXBE bit enables or disables the external bus.

Do not write 0 to this bit while a program is running from an external address space. Write 0 to this bit after access to the external bus is completed. Furthermore, when an external address space is included in the range of transfer by the bus masters other than the CPU (DMAC, DTC, EXDMAC, and EDMAC), prohibit DMA transfer before writing 0 to this bit.

After writing to the EXBE bit, confirm that its value has actually changed before proceeding with further processing.

When the EXBE bit is set to 1, the related I/O port settings must also be changed as required. For details, refer to section 23, Multi-Function Pin Controller (MPC).

3.2.3 System Control Register 1 (SYSCR1)

Address(es): 0008 0008h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	SBYRAME	ECCRAME	—	—	—	—	—	RAME
Value after reset:	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1

Bit	Symbol	Bit Name	Description	R/W
b0	RAME	RAM Enable	0: The RAM is disabled. 1: The RAM is enabled.	R/W
b5 to b1	—	Reserved	These bits are read as 1. The write value should be 1.	R/W
b6	ECCRAME	ECCRAM Enable	0: The ECCRAM is disabled. 1: The ECCRAM is enabled.	R/W
b7	SBYRAME	Standby RAM Enable	0: The standby RAM is disabled. 1: The standby RAM is enabled.	R/W
b15 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

RAME Bit (RAM Enable)

The RAME bit enables or disables the RAM.

A 0 should not be written to this bit during access to the RAM. When accessing the RAM immediately after changing the RAME bit from 0 to 1, make sure that the RAME bit is 1 before the access.

Even when the RAME bit is set to 0, the RAM retains its value. However, the voltage must be maintained at no less than the specified RAM hold voltage (VRAM), which is stipulated in section 63, Electrical Characteristics.

ECCRAME Bit (ECCRAM Enable)

The ECCRAME bit enables or disables the ECCRAM.

A 0 should not be written to this bit during access to the ECCRAM. When accessing the ECCRAM immediately after changing the ECCRAME bit from 0 to 1, make sure that the ECCRAME bit is 1 before the access.

Even when the ECCRAME bit is set to 0, the ECCRAM retains its value. However, the voltage must be maintained at no less than the specified RAM standby voltage (VRAM), which is stipulated in section 63, Electrical Characteristics.

SBYRAME Bit (Standby RAM Enable)

The SBYRAME bit enables or disables the standby RAM.

A 0 should not be written to this bit during access to the standby RAM. When accessing the standby RAM immediately after changing the SBYRAME bit from 0 to 1, make sure that the SBYRAME bit is 1 before the access.

Even when the SBYRAME bit is set to 0, the standby RAM retains its value. However, the voltage must be maintained at no less than the specified RAM hold voltage (VRAM), which is stipulated in section 63, Electrical Characteristics.

3.3 Details of Operating Modes

3.3.1 Single-Chip Mode

In single-chip mode, the external bus is disabled (SYSCR0.EXBE bit = 0) so that all I/O port pins are available for use as input or output port pins, inputs or outputs for peripheral functions, or as interrupt inputs.
If the high level is on the MD pin at the time of release from the reset state, the chip starts in single-chip mode. The on-chip ROM is enabled (SYSCR0.ROME bit = 1) at this time. The on-chip ROM can be disabled by software (by clearing the SYSCR0.ROME bit to 0), but it cannot be re-enabled (by setting the SYSCR0.ROME bit to 1) once this is done.
Setting the SYSCR0.EXBE bit to 1 (enabling the external bus) causes a transition to on-chip ROM enabled extended mode or to on-chip ROM disabled extended mode, making the external bus available.

3.3.2 On-Chip ROM Enabled Extended Mode

In this mode, the on-chip ROM is enabled (SYSCR0.ROME bit = 1) and the external bus extension is enabled (SYSCR0.EXBE bit = 1). This mode allows some I/O ports to be used as data bus input/output, address bus output, or bus control signal input/output. For details, refer to section 23, Multi-Function Pin Controller (MPC).
After the chip has started up in single-chip mode, setting the SYSCR0.EXBE bit to 1 (external bus enabled) causes it to make the transition to on-chip ROM enabled extended mode.
Writing 0 to the SYSCR0.EXBE bit (external bus disabled) causes a transition to single-chip mode (on-chip ROM enabled).
Writing 0 to the SYSCR0.ROME bit (on-chip ROM disabled) causes a transition to on-chip ROM disabled extended mode.

3.3.3 On-Chip ROM Disabled Extended Mode

In this mode, the on-chip ROM is disabled (SYSCR0.ROME bit = 0) and the external bus extension is enabled (SYSCR0.EXBE bit = 1). This mode allows some I/O ports to be used as data bus input/output, address bus output, or bus control signal input/output. For details, refer to section 23, Multi-Function Pin Controller (MPC).
After the chip has started up in single-chip mode, setting the SYSCR0.EXBE bit to 1 (external bus enabled) and setting the SYSCR0.ROME bit to 0 (on-chip ROM disabled) causes it to make the transition to on-chip ROM disabled extended mode.
In this mode, the on-chip ROM cannot be enabled by setting the SYSCR0.ROME bit to 1.
Writing 0 to the SYSCR0.EXBE bit (external bus disabled) causes a transition to single-chip mode (on-chip ROM disabled).

3.3.4 Boot Mode

In this mode, the on-chip flash memory modifying program (boot program) stored in a dedicated area within the MCU operates. The on-chip ROM (code flash memory and data flash memory) can be modified from outside the MCU by using a universal asynchronous receiver/transmitter (SCI1). For details, refer to section 62, Flash Memory (FLASH).
The chip starts up in boot mode if the low level is on both the MD and UB pins on release from the reset state.

3.3.5 USB Boot Mode

In this mode, the on-chip flash memory modifying program (boot program) stored in a dedicated area within the MCU operates. On-chip ROM (code flash memory and data flash memory) can be modified from outside the MCU by using the USB. For details, refer to section 62, Flash Memory (FLASH).
The chip starts up in boot mode (with the USB interface) if the low level is on the MD pin and the high level is on the UB pin on release from the reset state.

3.3.6 Boot Mode (FINE Interface)

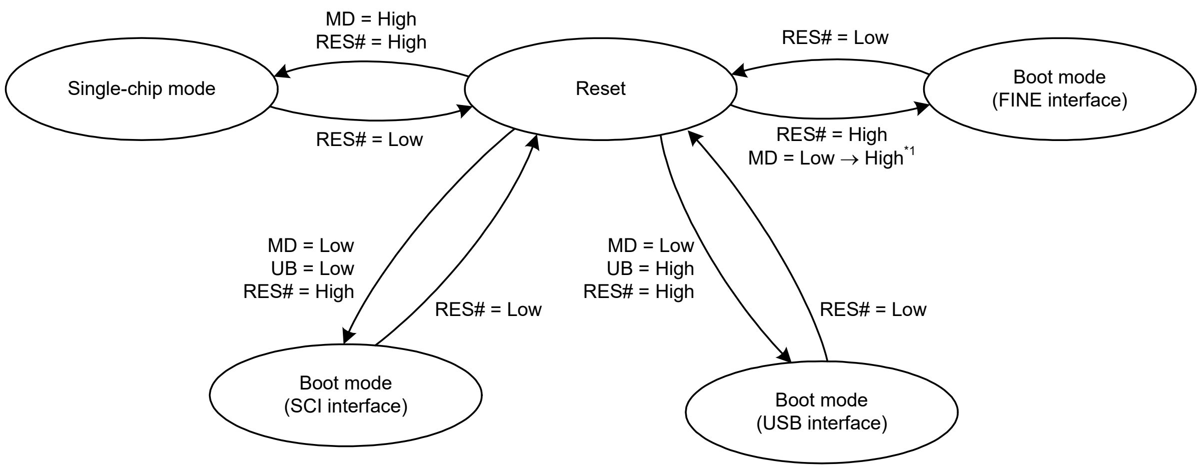
In this mode, the flash memory modifying program (boot program) stored in a dedicated area within the MCU operates. The on-chip ROM (code flash memory and data flash memory) can be modified from outside the MCU by using the FINE. For details, refer to section 62, Flash Memory (FLASH).

The chip starts up in boot mode (FINE interface) when the MD pin is set to the low level at the time of release from the reset state and then is then switched to the high level within 20 to 100 msec.

3.4 Transitions of Operating Modes

3.4.1 Operating Mode Transitions Determined by the Mode-Setting Pins

Figure 3.1 shows operating mode transitions determined by the settings of the MD pin and the UB pin.



Note 1. After release from the reset state, switch the MD pin to high level within 20 to 100 msec.

Figure 3.1 Mode-Setting Pin Level and Operating Mode

3.4.2 Operating Mode Transitions According to Register Setting

Figure 3.2 shows operating mode transitions according to the setting of the ROME and EXBE bits in SYSCR0.

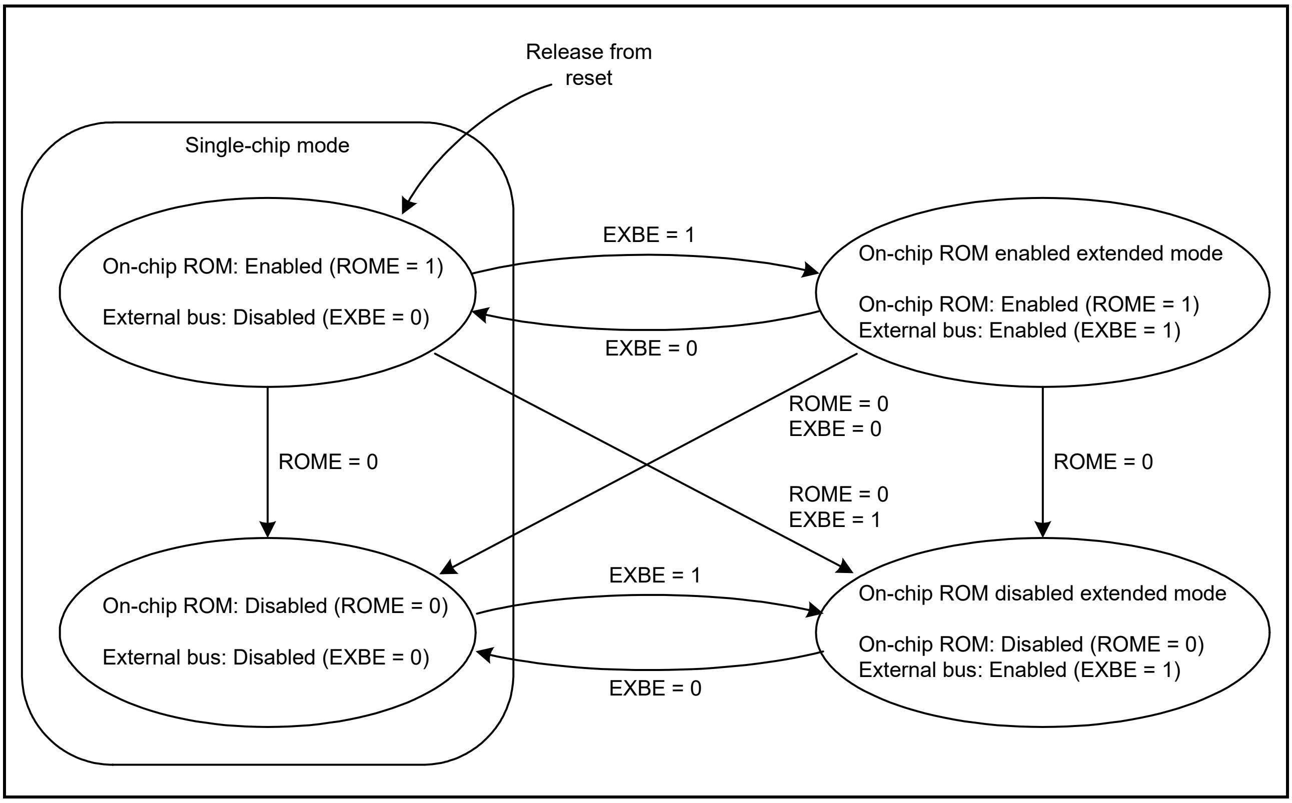


Figure 3.2 Setting of SYSCR0.ROME and EXBE Bits and Operating Modes